CHAPTER 12

WATER RESOURCES

CHAPTER ESSENCE AT A GLANCE:
Analyze India freshwater availability (surface and groundwater), overwhelming sectoral demand in agriculture, water depletion in north-western states, water quality degradation, watershed management models (Haryali, Neeru-Meeru, Arvary Pani Sansad), and rainwater harvesting.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Freshwater Availability and Basins:** India possesses 4 percent of world water resources to sustain 18 percent of world population. Total annual precipitation is about 4,000 cubic km. Replenishable groundwater is 432 billion cubic metres (BCM), heavily concentrated in the Ganga-Brahmaputra basins.

• **Overwhelming Agricultural Water Demand:** Agriculture accounts for 89 percent of surface water and 92 percent of groundwater utilization, driven by intensive irrigation of water-demanding crops like rice and sugarcane in semi-arid states.

• **Over-Exploitation and Chemical Contamination:** Unregulated tubewell over-draft in Punjab, Haryana, and Western UP causes precipitous water table drops, severe fluoride contamination in Rajasthan, and toxic arsenic poisoning in the Ganga delta.

• **Watershed Management and Rainwater Harvesting:** Watershed development integrates conservation of water, soil, and vegetation. Exemplars: Haryali (Central Government), Neeru-Meeru (Andhra Pradesh), and Arvary Pani Sansad (Alwar, Rajasthan). Rainwater harvesting recharges depleted aquifers.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$\text{Total Replenishable Groundwater} \approx 432 \text{ Billion Cubic Metres (BCM) per Year}$$

National Replenishable Groundwater Endowment

$$\text{Sectoral Surface Water Utilization} \iff \text{Agriculture } (89\%) + \text{Domestic } (9\%) + \text{Industrial } (2\%)$$

Surface Water Utilization Breakdown

$$\text{Watershed Management Watersheds} \to \{\text{Haryali (Central)}, \text{Neeru-Meeru (AP)}, \text{Arvary Pani Sansad (Alwar)}\}$$

Prominent Indian Watershed Management Programmes

$$\text{Rainwater Harvesting (RWH)} \iff \text{Rooftop Surface Collection} \to \text{Recharge Pits and Borewells}$$

Rooftop Aquifer Recharge Methodology

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Never claim that industrial or domestic use dominates water demand in India; agriculture accounts for roughly 90 percent of both surface and groundwater consumption.

• **Alert:** Excessive groundwater extraction causes toxic arsenic contamination in West Bengal and Bihar, whereas in Rajasthan and Andhra Pradesh it leads to dangerous fluoride contamination.

• **Alert:** Neeru-Meeru is a participatory watershed management programme in Andhra Pradesh, while Arvary Pani Sansad was built by villagers in Alwar, Rajasthan.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Explain why agriculture accounts for the overwhelming majority of water utilization in India. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **High Crop Water Demands:** Dominant crops like paddy, sugarcane, and jute require massive, sustained irrigation water across semi-arid plains. **[1.5 Marks]**

• **Spatial and Temporal Rainfall Deficit:** Most of India annual rainfall occurs within barely three to four monsoon months, necessitating round-the-year artificial irrigation for winter and summer crops. **[1.5 Marks]**

2. Discuss the consequences of excessive over-utilization of groundwater in north-western states like Punjab and Haryana. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Water Table Plunge:** Heavy tubewell extraction has caused the underground water table to drop precipitously, drying shallow borewells. **[1 Mark]**

• **Salinization and Waterlogging:** Unregulated canal and tubewell flood irrigation in poorly drained soils leads to soil alkalization and surface crusting. **[1 Mark]**

• **Rising Pumping Costs:** Farmers must drill deeper wells and install higher horsepower submersible pumps, escalating agrarian debt. **[1 Mark]**

3. What is watershed management? Explain any two prominent watershed development programmes in India. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Definition:** The holistic, scientific management and conservation of surface and groundwater resources, soil, and vegetation within a natural watershed drainage basin. **[1.5 Marks]**

• **1. Haryali:** A Central Government sponsored programme implemented by Gram Panchayats enabling rural populations to harvest rainwater for drinking, irrigation, and fisheries. **[1.5 Marks]**

• **2. Neeru-Meeru (Water and You):** A community-based watershed programme in Andhra Pradesh that constructed percolation tanks, check dams, and contour trenches. **[1 Mark]**

4. Describe the objectives and methods of rainwater harvesting in drought-prone areas. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Key Objectives:** Prevent surface runoff loss, replenish depleted groundwater aquifers, reduce soil erosion, and secure local water supply during dry seasons. **[1.5 Marks]**

• **Practical Methods:** Rooftop rainwater catchment routed through filters into recharge pits, dugwells, borewells, check dams, and local percolation tanks. **[1.5 Marks]**

5. Name the states where groundwater depletion has caused toxic concentrations of: (a) Arsenic, and (b) Fluoride. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **(a) Toxic Arsenic Contamination:** West Bengal, Bihar, and parts of the Ganga plain. **[1.5 Marks]**

• **(b) Toxic Fluoride Contamination:** Rajasthan, Gujarat, Andhra Pradesh, and Maharashtra. **[1.5 Marks]**

6. Highlight any three key priorities laid down in the National Water Policy of India. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **1. Drinking Water Priority:** Allocates the highest first-order priority to safe drinking water for all human beings and livestock. **[1 Mark]**

• **2. Irrigation and Efficiency:** Mandates modernization of canal networks and the adoption of water-conserving drip and sprinkler technologies. **[1 Mark]**

• **3. Ecological and Quality Safeguards:** Demands stringent monitoring of industrial effluent discharges to maintain ecological minimum river flows. **[1 Mark]**

7. Why is the Ganga-Brahmaputra basin characterized by very high water resources potential compared to peninsular river basins? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Perennial Snowmelt:** The Ganga and Brahmaputra originate in Himalayan glaciers and receive continuous snowmelt in summer supplemented by torrential monsoon rain. **[1 Mark]**

• **Rainfall Volume and Relief:** The catchment receives heavy precipitation over enormous catchment basins, unlike the seasonal, rain-dependent peninsular rivers. **[1 Mark]**